

Digital Twin for Surface Treatment Industries

Mini project demonstrating DES + optimization coupling for finite capacity production planning

DES Model
SimPy-based workshop simulation

Optimization
Scheduling and batch grouping

Decision Support
Timeline and performance metrics

Research context: Digital Twin for Short-Term Planning, IMT Mines Albi / INFINITY project

Project Overview

What the mini project demonstrates

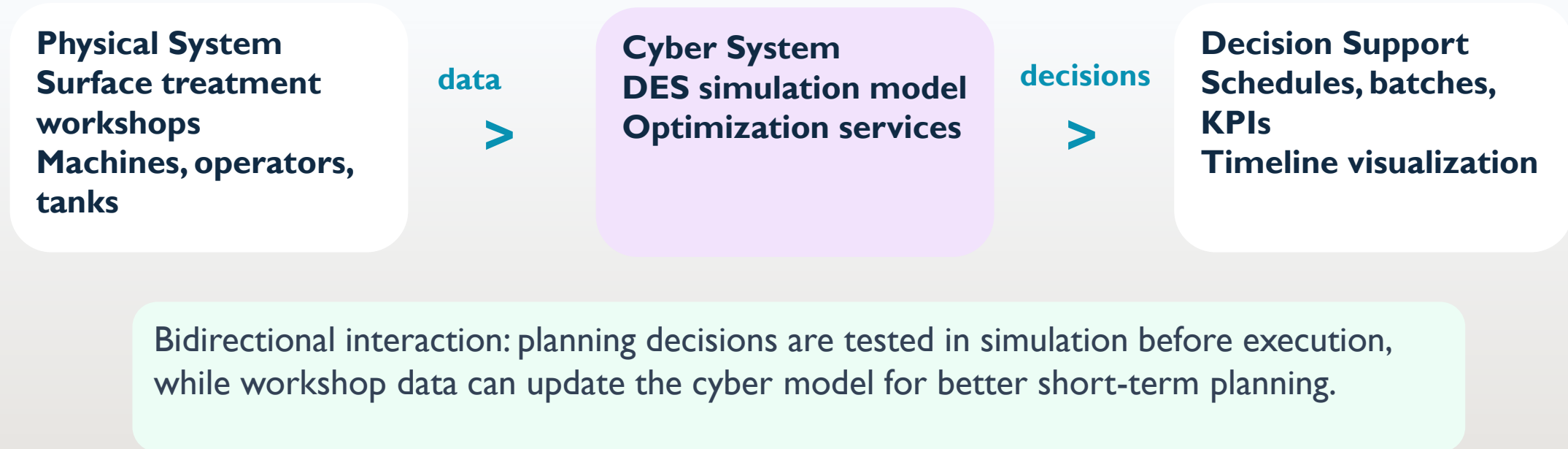
- A digital twin system for short-term production planning in surface treatment workshops.
- Discrete Event Simulation models workshop execution, queues, resources and order flow.
- Optimization services support scheduling, sequencing and batch grouping decisions.
- Finite capacity planning coordinates multiple workshops with resource constraints.
- Dashboards and KPIs support decision-making for planners and supervisors.

Industry Context

- Made-to-order production
- Short promised lead times
- High customer diversity
- Variable product mix
- Manual + automated processes

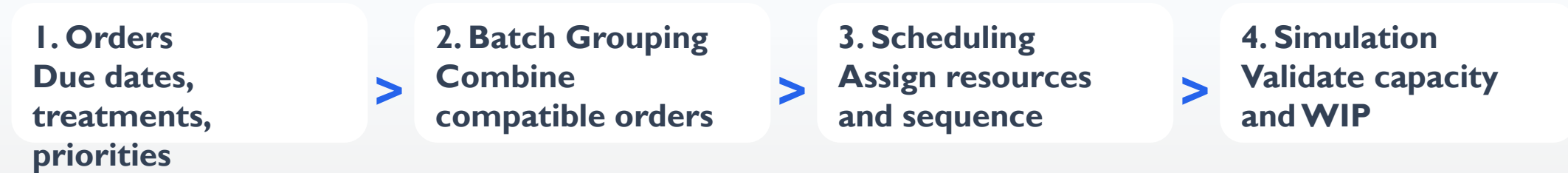
Digital Twin Architecture

Physical system connected to cyber planning services



Finite Capacity Planning Methodology

Coupling simulation and optimization



- Multi-stage surface treatment process with machine, operator and material constraints.
- Short-term planning horizon: days; medium-term horizon: weeks.
- Performance indicators: lateness, makespan, WIP, utilization and throughput.
- The optimization proposes a plan; DES checks whether it is feasible under finite capacity.

Implementation and Technologies

Project structure and technical stack

Project Modules

src/simulation

- workshop.py
- order.py
- resources.py

src/optimization

- scheduler.py
- batch_grouping.py

src/visualization

- dashboards.py

Technology Stack

Simulation: SimPy

Optimization: NumPy / SciPy

Data analysis: Pandas

Visualization: Matplotlib / Plotly

Configuration:YAML

Language: Python 3.8+

Included outputs: workshop simulation framework, order and batch management, optimization skeleton, example scenarios and performance metrics.

Conclusion and Future Extensions

How the mini project supports PhD work

- The project demonstrates a practical digital twin architecture for surface treatment planning.
- DES captures workshop behavior under finite capacity and operational constraints.
- Optimization services provide scheduling and batch grouping recommendations.
- The framework can support planner decisions through KPIs and timeline visualization.

Future Extensions

- Reinforcement learning for adaptive scheduling
- Real-time data collection interface
- Database integration for order history
- Web dashboard for planners
- Advanced optimization algorithms

Final message: this mini project links simulation, optimization and decision support for resilient short-term planning in surface treatment industries.

